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ARTICLE

Ultralarge Flakes of $Ti_3C_2T_x$ MXene *via* Soft Delamination

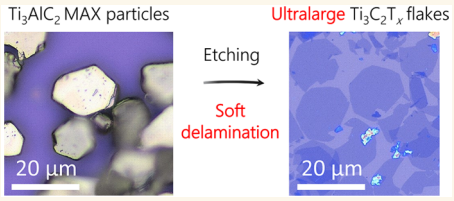
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ABSTRACT: Two-dimensional (2D) titanium carbide MXene ($Ti_3C_2T_x$) has attracted significant attention due to its combination of properties and great promise for various applications. The size of the 2D sheets is a critical parameter affecting multiple properties of assembled films, fibers and 3D structures. The increased lateral size of MXene flakes can benefit not only their assemblies by improving the interflake contacts and alignment but also fundamental studies at the individual flake level, allowing for facile patterning and investigation of intrinsic physical properties of MXenes. Increasing the average size of the parent MAX phase is one of the strategies previously used to increase the flake size of the resultant MXene. Here, we show that the protocol used for the next step of the synthesis procedure, delamination of multilayer MXene into individual nanosheets, significantly affects the lateral size of the resultant flakes. We developed a soft delamination approach, which prevents fracture of flakes and preserves their size. Combining this approach with the large-grain Ti_3AlC_2 MAX phase precursor, we achieved individual flakes of up to 40 μm in lateral size. These flakes can be used for patterning multiple contacts and fabrication of field-effect transistors for multiprobe electrical characterization and other measurements. These findings indicate the importance of controlling the delamination process in order to achieve large MXene flakes and improve properties of MXene-based materials and devices.



KEYWORDS: *MXene, delamination, large flakes, Ti_3C_2, 2D materials*

INTRODUCTION

Starting with the first exfoliation of graphene crystals,[1] different two-dimensional (2D) materials, including hexagonal boron nitride and transition metal chalcogenides, have been isolated and investigated.[2] Fabrication techniques, such as micromechanical exfoliation and chemical vapor deposition (CVD),[3] enabled fabrication of large 2D crystals, providing a facile platform to study fundamental properties of 2D materials and heterostructures. In comparison, synthesis of 2D transition metal carbides and/or nitrides (MXenes) *via* chemical etching has not provided such large individual crystals, limiting the studies at the single-flake level to just a few examples.[4−12] Large flakes are also extremely important for bulk applications of MXenes in printed electronics, energy storage, and composite materials,[13,14] all of which are rapidly advancing due to the scalability of the solution synthesis of MXenes and their facile processing. Average flake size is widely recognized as one of the critical parameters for any solution-synthesized 2D material. For example, preparation of ultralarge flakes of another popular solution-processable 2D material, graphene oxide (GO), had been an important target of many dedicated studies.[15,16] Besides GO, fabrication of large flakes of other 2D materials in solution had been challenging, keeping the typical flake size of oxides and chalcogenides at the micrometer level.[17]

Titanium carbide ($Ti_3C_2T_x$) was the first discovered[18] and remains the most studied MXene[14] with many potential applications facilitated by a combination of its physical and chemical properties.[14,19] High electrical conductivity,[20] easy dispersion in water,[21] and rich surface chemistry[22] are among the advantages of MXenes compared to other 2D materials, which brings MXenes to the forefront of research and development in the materials field. In case of $Ti_3C_2T_x$ MXene, its original synthesis heavily relied on sonication of the chemically etched Ti_3AlC_2 MAX phase, and the reported flake size was typically in the sub-micrometer range.[18] Further synthetic improvements resulted in the minimally intensive layer delamination (MILD) method, which avoided the extensive sonication of etched product, and the size of

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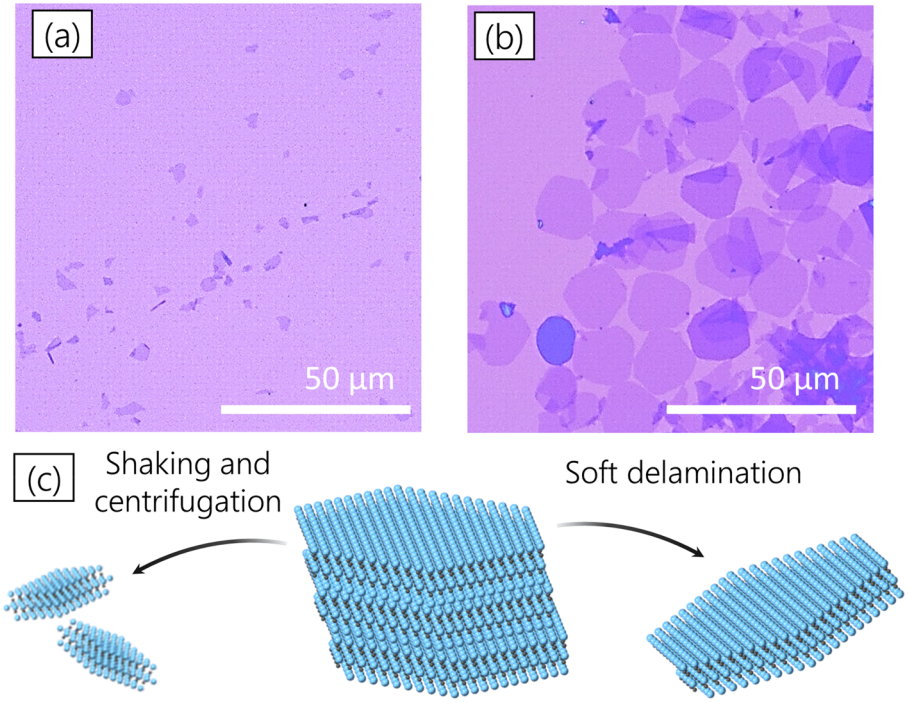


Figure 1. Comparison of single-layer MXene flakes prepared by regular delamination and soft delamination. Optical micrographs of the flakes on a Si/SiO_2 wafer prepared by (a) traditional delamination approach and (b) soft delamination. (c) Schematic illustration of the two approaches.

$Ti_3C_2T_x$ MXene flakes increased up to several micrometers.[5] This work reports on an improvement in the $Ti_3C_2T_x$ flake size by another order of magnitude, which will benefit a broad range of MXene applications.

The lateral size of the nanosheets plays a crucial role in the resultant properties. Since MXenes are typically used as assembled films or fibers, the flake size significantly affects the interflake contacts, which are important for conductivity, mechanical, and thermal properties. Several publications[23,24] have demonstrated increased conductivity of vacuum-filtered films and assembled fibers with increasing flake size of $Ti_3C_2T_x$ due to improved interflake contacts and flake alignment. Zhang *et al.*[25] also demonstrated that the parent MAX phase used for the etching process affects the size of the resultant MXene flakes. By eliminating the small MAX particles before the etching step, the authors prepared MXene with increased flake size, which translated into enhanced mechanical properties and electrical conductivity of the assembled films. Larger $Ti_3C_2T_x$ MXene flakes should also translate into improved barrier properties in membranes as well as more efficient percolation of the flakes in a polymer or ceramic matrix. Still, translating progress in the fabrication of large MAX phase particles[26,27] into large MXene flakes had been challenging.

While MXenes are typically used as multiflake assemblies, single-flake characterization and property measurements are crucially important and require a large flake size, ideally over 10−20 μm. Recent studies have suggested that single-layer MXene flakes can be used as electrodes for various electronic devices[28] or high-voltage interconnects,[11] due to the high electrical conductivity and breakdown current density of $Ti_3C_2T_x$. In comparison to graphene or transition metal dichalcogenides, where large-area monolayers can be prepared *via* CVD or micromechanical exfoliation,[3] MXene flakes used in single-flake studies are limited in size due to the nature of the synthetic process.

Here we demonstrate that delamination is the size-limiting step for MXene synthesis. Particularly, shaking, which is typically employed for "mild" delamination, negatively affects the size by breaking the flakes. We developed a soft delamination approach that preserves the flakes while allowing them to separate from the multilayer particles when deposited on substrates. Using this approach, we demonstrate the synthesis of ultralarge MXene flakes with a lateral size of up to 40 μm. The large size of individual flakes is expected to translate into improved electrical, mechanical, and barrier properties of MXene films and composites. It also provides an opportunity for patterning multiple electrical contacts on a single flake, as shown with a field-effect transistor, which is beneficial for electrical measurements of MXenes. Our findings indicate parameters necessary to control to fabricate large MXene flakes, which will enable the production of MXene films and assemblies with improved properties.

RESULTS AND DISCUSSION

In general, $Ti_3C_2T_x$ MXene is prepared by acid etching of aluminum from the Ti_3AlC_2 MAX phase.[14] After the etching is finished and the multilayer MXene is washed to neutral pH, the delamination process is typically carried out by stirring the powder in a LiCl solution, followed by repeated cycles of shaking and centrifugation.[26] While some flakes can exceed 10 μm in size, on average, the flakes are smaller (1−10 μm as measured by DLS) than the parent MAX phase particles and multilayer MXene, showing a wide distribution of sizes. Furthermore, these flakes often demonstrate rough edges and irregular shapes that are different from the shape of the precursor MAX particles.[26] We assumed that the delamination step was causing the individual MXene flakes to break, thus

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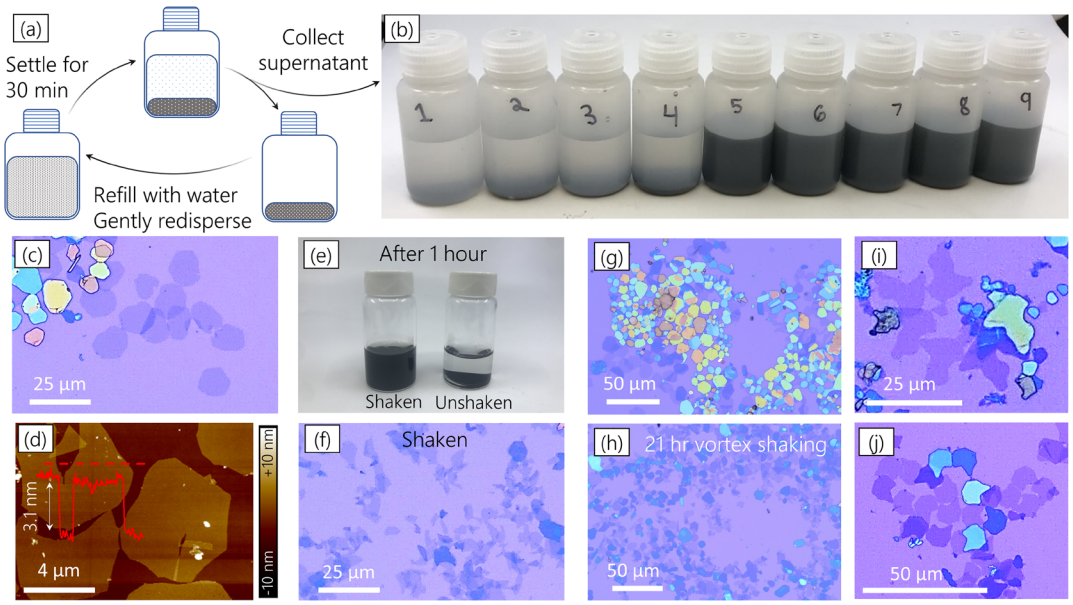


Figure 2. Soft delamination of MXene. (a) Schematic of the soft delamination process. (b) Photograph of the bottles with MXene solutions collected during delamination. (c) Optical micrograph of the flakes deposited on the Si/SiO_2 substrate *via* drop-casting. (d) AFM image of delaminated flakes. The inset shows the height profile along the dashed line. (e) Photograph of vials before and after hand shaking. (f) Optical micrograph of the flakes from the shaken solution deposited on a Si/SiO_2 substrate. (g, h) Optical micrographs of flakes deposited from an unshaken solution and the solution shaken with vortex. (i, j) Optical micrographs of flakes delaminated from the multilayer particles.

resulting in a decreased flake size and irregular shape. Figure 1a shows a typical optical microscopy image of $Ti_3C_2T_x$ flakes prepared by the traditional delamination protocol.

In this report, we demonstrate that by utilizing a soft delamination approach it is possible to preserve the flakes and translate the size of the parent MAX phase particles into individual delaminated flakes. An example of the flakes prepared by this protocol is shown in Figure 1b. These flakes demonstrate crystallographic edges with the overall shapes resembling the parent MAX particles (Figure S1a,b). All the light-colored flakes are of the same thickness and represent single-layer MXene. There are a few double- and multilayer flakes in the view field in Figure 1b, which are darker in color.

The protocol for the soft delamination is schematically shown in Figure 2a. After stirring the multilayer particles in LiCl solution, the solution was allowed to settle for approximately 30 min. During this time, most of the particles precipitate to the bottom of the container. After that, the supernatant was carefully pipetted out of the container and collected. The bottle was then refilled with water, and MXene was gently redispersed by inverting the bottle several times. After allowing the solution to settle for about 30 min, the procedure was repeated multiple times until the supernatant became transparent, indicating a low concentration of MXene.

Figure 2b shows the bottles containing the solution collected during this process. At the beginning (bottles 1−4), the collected particles settled quickly due to the presence of concentrated LiCl in solution. As the concentration of LiCl decreases, MXene flakes start to delaminate spontaneously, forming a stable colloidal solution, which visually appears much darker. The most concentrated solution was found to contain approximately 2.5 mg of MXene per milliliter, with the total yield of approximately 47%.

Figure 2c shows optical microscopy image of the flakes from bottle #5 deposited on a Si/SiO_2 substrate. Optical microscopy is the best tool to evaluate the delamination process, as it is quick and convenient and provides excellent contrast to image even single-layer MXene flakes.[29] Furthermore, optical microscopy can easily distinguish the single-, double-, and few-layer flakes. In comparison, scanning electron microscopy (SEM) imaging on Si/SiO_2 does not provide enough contrast to properly visualize the flakes (SE2 detector, Figure S1d) or distinguish single- and few-layer flakes (InLens detector, Figure S1c). Anodic aluminum oxide (AAO) membranes, which are typically used for SEM imaging of MXene flakes,[30] can provide the necessary contrast to identify the single-layer flakes; however, we found that the porous nature of the membranes may damage the large flakes upon drying/deposition due to capillary forces in the membrane channels (Figure S1e,f). Transmission electron microscopy (TEM) imaging is relatively time- and cost-consuming and has a limitation of the small field of view (Figure S1g,h), making this technique inconvenient for evaluating large flakes. Distinguishing the number of layers is difficult in both electron microscopy techniques. Thus, we primarily used optical microscopy as an imaging tool to assess delamination.

The deposited MXene flakes (Figure 2c) show a mixture of few- and multilayer MXene particles alongside the single-layer flakes. These flakes are larger than those prepared *via* the conventional delamination protocol (Figure 1a), with an average size of approximately 14 μm. The size distribution for the single-layer MXene flakes is shown in Figure S2. Furthermore, the shape of the monolayer flakes resembles the shape of the multilayer particles and the precursor MAX particles (Figure S1a,b),[26] indicating that the soft delamination technique preserves the flake size and shape. Atomic force microscopy (AFM) imaging of these flakes (Figure 2d) further confirms that the flakes are single-layer. The height profile was measured at approximately 3.1 nm, which exceeds the theoretical monolayer MXene thickness due to trapped water

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and other molecules in between the flake and the substrate, but is in agreement with the previous measurements of monolayer $Ti_3C_2T_x$ MXene on Si/SiO_2.[5,6]

To illustrate the necessity for soft delamination, we hand shook a vial of MXene with the large flakes. After shaking the solution, MXene immediately formed a stable colloidal solution (Figure 2e), which is typically observed during MXene synthesis.[30] The shaking, however, breaks the flakes into smaller particles, as evident from the optical microscopy images of the resultant MXene (Figure 2f). This experiment clearly illustrates that the large flakes, having an aspect ratio of over 10 000, are fragile, and the soft delamination is necessary to preserve the flake size. We speculated that applying low-intensity and well-controlled shear force *via* swirling the solution (using vortex or orbital shaking) or spin-coating at low speeds can aid the delamination process. Indeed, vortex shaking for 21 h significantly reduced the number of multilayer particles (compare Figure 2g and h). Spin-coating of the solution on Si/SiO_2 substrates at low speeds (500 rpm) yields similar results, reducing the number of multilayer particles observed (Figure S3). However, few-layer particles and broken single-layer flakes were still observed, and further optimization of the shaking and deposition parameters is necessary to produce uniformly single-layer MXene.

Interestingly, single-layer flakes usually concentrate close together in several areas on the substrate, while other sites typically consist of only multilayer particles. On multiple occasions, we also observed areas where single-layer MXene flakes were in close proximity to the parent multilayer particles. Figure 2i,j demonstrates examples of MXene flakes delaminated from the parent multilayer particle in the same area, as the specific shape of the multilayer particle is inherited by the single-layer flakes. This fact suggests that the single-layer MXene flakes are not present in solution by themselves but most probably are delaminated from the multilayer particles during processing as the capillary or created shear forces separate the flakes.

To confirm this mechanism, we performed optical microscopy analysis of MXene in solution while allowing water to evaporate. Video recording of the process is available as Supplementary Video 1; the time-lapse sequence of images is shown in Figure 3. In this case, a sample of large MXene particles, the synthesis of which will be discussed later, was delaminated using the soft delamination protocol, and a droplet of the solution was placed on a Si/SiO_2 substrate. In solution, swelled multilayer MXene particles are elongated along the direction perpendicular to the basal plane of the nanosheets, indicating significant separation of MXene layers. Similar swelling of etched MAX was recently observed as a result of osmotic swelling.[31] Flakes in these structures, however, still have several points of connection. Upon drying, the multilayer MXene particles move and MXene sheets separate from the parent multilayer particles at the points where two larger stacks break apart (indicated by arrows in Figure 3). Eventually, these flakes deposit in the nearby area (indicated by circles in Figure 3) when the water is completely evaporated. This process is not uniform, as some multilayer particles do not separate into individual flakes, yielding multilayer MXene deposits on the substrate (stack 1 in Figure 3). In some cases, the flakes continue to have some points of contact throughout the drying process, resulting in elongated ribbons of overlapping flakes (parts of the stack 2 in Figure 3). This mechanism was only observed for freshly deposited

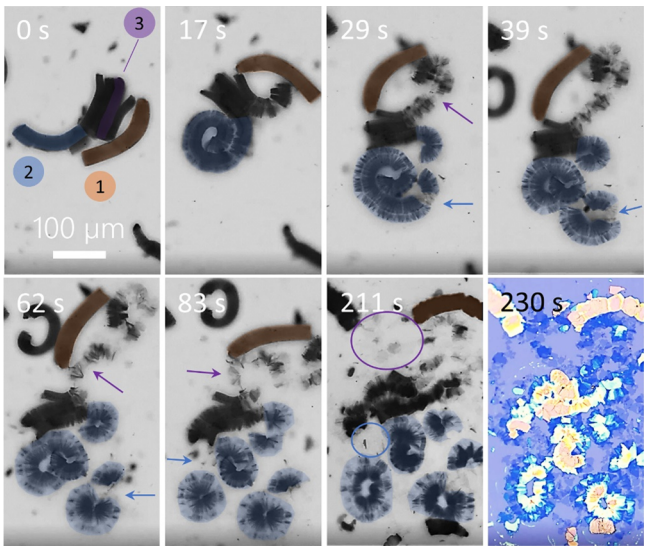


Figure 3. Time-lapse sequence of optical micrographs of multilayer MXene flakes suspended in a water droplet deposited on a Si/SiO_2 substrate. Upon drying, the flakes separate from the parent multilayer MXene (indicated by arrows) and deposit on the substrate (indicated by circles). The last image (230 s) shows a completely dried sample. The full video is available as Supplementary Video 1.

samples. For the samples that were stored in solution for an extended period of time (approximately one month), the single-layer sheets were already separated, and the multilayer particles that were left in solution did not delaminate during the drying process. As a result, single- and few-layer MXene flakes were randomly distributed across the substrate (Figure S4) when deposited *via* drop-casting. These experiments indicate that the soft delamination process is not instantaneous, as the freshly deposited flakes are still in the process of delamination, and aging or agitation is required to separate individual flakes from the multilayer particles. Importantly, these experiments demonstrate another advantage of the large flakes fabricated in this work: the ability to perform imaging using optical microscopy and visualize the delamination process *in situ*.

The size of the resultant MXene flakes is limited by the size of the parent MAX particles. In order to increase the size of MXene flakes, we prepared larger MAX particles by annealing the precursor powder containing TiC, Ti, and Al for 24 h at 1400 °C, compared to the usual 2 h (see the Experimental Section for details). Figure 4a shows the powder X-ray diffraction (XRD) pattern of the MAX powder, which is in perfect agreement with the XRD of Ti_3AlC_2 obtained after annealing for 2 h. The resultant powder was sieved into three fractions with the particle sizes of <20, 20−38, and 38−75 μm, which accounted for 30, 55, and 15 wt %, respectively. In comparison, 80 wt % of the MAX powder prepared by annealing for 2 h was particles below 20 μm in size.

We selected two of the largest fractions, namely, 20−38 and 38−75 μm, and etched them for different periods of time (see the Experimental Section for details). In both cases, etching for 72 h provided complete etching with no MAX phase detected *via* XRD; only the (00*l*) series of MXene peaks was observed, starting with (002) peak at $2\theta = 8.5°$. In contrast, 48 and 24 h etching procedures showed the peaks of residual Ti_3AlC_2,

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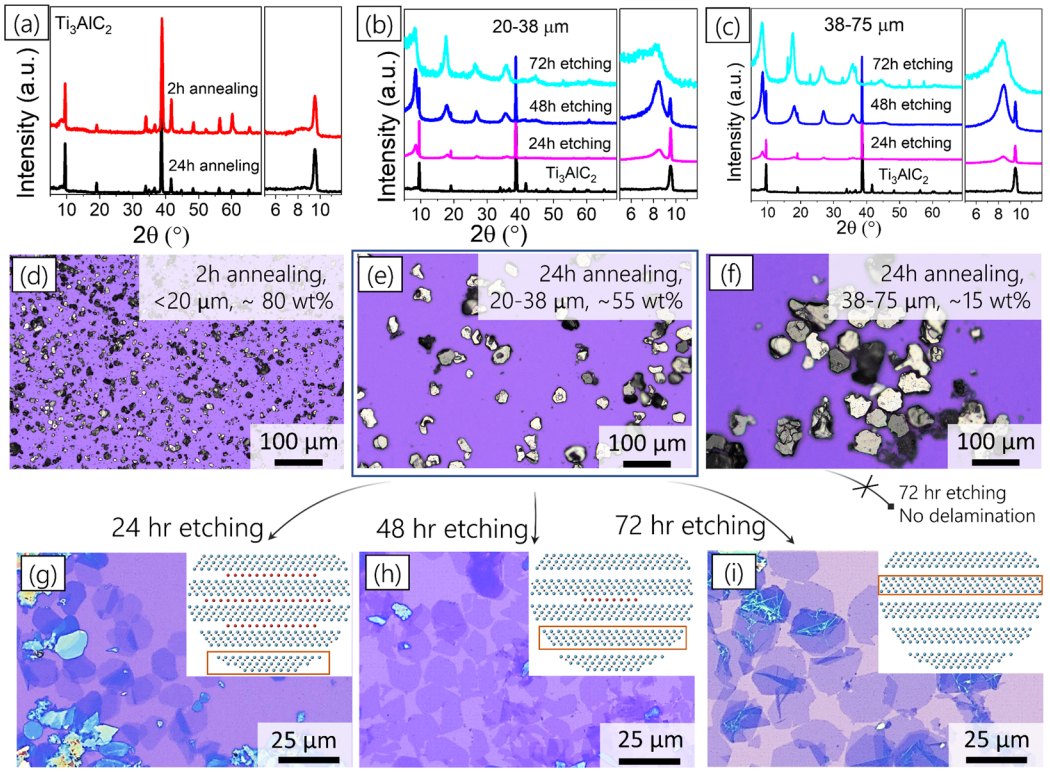


Figure 4. Synthesis of large MXene flakes from large MAX particles. (a) XRD of Ti_3AlC_2 MAX phase prepared by annealing at 1400 °C for 2 or 24 h. (b, c) XRD of MXene etched for different time periods. (d, e, f) Optical micrographs of MAX phase. (g, h, i) Optical micrographs of MXene etched from the 20−38 μm MAX fraction for different time periods, delaminated using the soft delamination protocol and deposited on Si/SiO_2 substrates. Insets schematically demonstrate the etching process that starts from outer layers, which leads to different sizes of the flakes.

which decreased in intensity as the etching time increased (Figure 4b,c).

Figure 4d shows optical microscopy images of the MAX particles prepared by annealing for 2 h, compared to the particles from the 20−38 and 38−75 μm fractions (Figure 4e,f). The longer-annealed particles are noticeably larger, which increases the limit for the possible MXene flake size. The size distribution for the MAX particles is shown in Figure S2a−d. Unfortunately, our attempts to delaminate the largest 38−75 μm fraction etched for 72 h were unsuccessful; MXene flakes did not spontaneously delaminate, and we observed only multilayer MXene particles as a result. The reasons are not clear at the moment, but may include strong bonding between large MXene sheets, for instance, due to incompletely removed etching byproducts.

However, delamination of the 20−38 μm particles using the soft delamination protocol was successful. Figure 4g,h,i shows optical microscopy of the flakes obtained after soft delamination of the 20−38 μm fraction etched for different times. Interestingly, even incomplete etching for 24 and 48 h yielded single-layer flakes, though the flake size was smaller. The size distribution for the MXene flakes is shown in Figure S2f−h. This fact can be rationalized by examining the mechanism of the etching process described in ref 32. The etching front is not moving uniformly from the edge of the MAX particles but predominantly starts from the surface layers. As a result, in the case of incomplete etching, where the MAX phase is still present in the sample, the top layers of the MAX particles can be completely etched and provide single-layer flakes upon delamination. As the etching process continues, deeper layers of the MAX phase are transformed into MXene and become available for delamination, providing larger MXene flakes. This mechanism is schematically illustrated in the insets of Figure 4g,h,i. Naturally, as the MAX phase used in these experiments was sieved below 38 μm, the largest flakes we observed were approximately 40 μm in size, as shown in Figure S5.

The large average size of MXene flakes achieved in this study expands the horizons of their electronic applications.[33] Many recent studies on electronic,[34,35] photovoltaic,[36] and sensing[37,38] applications of MXenes employed thin films consisting of multiple overlapping flakes. Noteworthy, the electrical properties of such films are affected by interflake contacts, which strongly depend on the surface chemistry of MXenes and the presence of intercalated molecules.[39] These factors greatly increase the number of variables affecting the device performance. However, with flake dimensions well exceeding 10 μm, as demonstrated in this work, it is feasible to consider individual MXene flakes as electrical contacts[28] or active channels[5,10,12] of various electronic devices. Importantly, such large flakes provide sufficient space for facile patterning of multiple electrical contacts, as we demonstrate for the MXene device shown in Figure 5a. The SEM image in this figure shows a monolayer $Ti_3C_2T_x$ flake with six Cr/Au electrodes patterned by standard electron beam lithography (EBL). In general, such electrode geometry enables several kinds of electrical characterization ranging from four-probe testing to Hall effect measurements. With the size of this flake of >10 μm, additional metal probes could have been placed on this MXene monolayer, if necessary.

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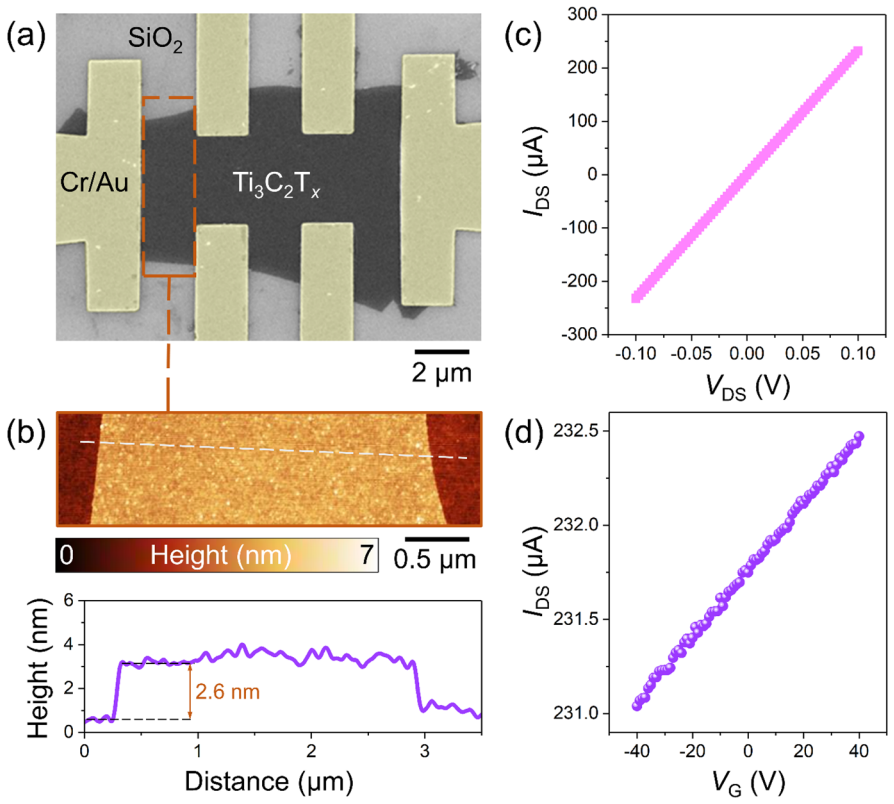


Figure 5. Electrical measurements of monolayer $Ti_3C_2T_x$ flakes. (a) False-color SEM image of a multiterminal device based on a large monolayer $Ti_3C_2T_x$ MXene flake. The Cr/Au electrodes are colored in yellow for clarity. (b) Top panel: AFM image of a portion of the $Ti_3C_2T_x$ MXene device channel that is marked by a rectangle in panel (a). Bottom panel: AFM height profile measured across the $Ti_3C_2T_x$ MXene flake along the gray dashed line in the top panel. (c) I_{DS}–V_{DS} dependence measured between the inner probes of the $Ti_3C_2T_x$ device shown in (a). (d) I_{DS}–V_G dependence measured between the inner probes of the $Ti_3C_2T_x$ device shown in (a) at V_{DS} = 0.1 V.

Figure 5b shows an AFM image of a portion of the $Ti_3C_2T_x$ device channel. Even after the device fabrication, the flake has a clean and uniform surface without visible EBL resist residues. The height profile of this flake reveals its thickness of about 2.6 nm, which is consistent with the values previously measured by AFM for monolayer $Ti_3C_2T_x$ flakes on Si/SiO_2 substrates.[5,6,11] It should be noted that while the theoretical thickness of a $Ti_3C_2T_x$ monolayer is about 0.98 nm,[6] in the experimental AFM images, the monolayer $Ti_3C_2T_x$ flakes on Si/SiO_2 appear much thicker, with thicknesses typically in the range of 2.5–3 nm, which is likely due to the effect of adsorbed molecules, such as water.

The $Ti_3C_2T_x$ MXene flakes prepared in this study were highly electrically conductive, as illustrated by Figure 5c. The figure shows the current (I_{DS})–voltage (V_{DS}) dependence that was measured between the inner probes of the device shown in Figure 5a. The I_{DS}–V_{DS} dependencies exhibit an ohmic behavior, suggesting good electrical contacts between the $Ti_3C_2T_x$ MXene and the Cr/Au electrodes. From the I_{DS}–V_{DS} data, the geometrical parameters of the device, and the theoretical thickness of the $Ti_3C_2T_x$ monolayer of 0.98 nm,[6] we calculated the electrical conductivity of about 6500 S cm^{-1}, which is consistent with the recently reported measurements[11] of MXenes prepared by the same synthetic method.[26] Because the device was prepared on a p-doped Si substrate covered with a 300-nm-thick layer of SiO_2, we could use the electrically conductive silicon layer as a global gate electrode in field-effect measurements. The resulting dependence of the current between the inner probes in the same device (Figure 5a) on the gate voltage (V_G) is shown in Figure 5d. The positive slope of the I_{DS}–V_G dependence suggests the n-type conductivity of $Ti_3C_2T_x$ MXene, which is consistent with the previous measurements.[5,11] From the I_{DS}–V_G dependence, we also calculated the field-effect electron mobility for monolayer $Ti_3C_2T_x$ MXene; see details in our previous works.[5,11] The calculated field-effect electron mobility of about 4.2 cm^2 V^{-1} s^{-1} for the device shown in Figure 5a is consistent with the values previously reported for monolayer $Ti_3C_2T_x$ MXene[11] prepared by the same synthetic method.[26]

CONCLUSIONS

The size of the delaminated $Ti_3C_2T_x$ MXene depends on the Ti_3AlC_2 precursor particles, synthesis approach, and delamination protocol. We show that sonication and shaking decrease the average size of MXene flakes, causing the flakes to break into smaller particles. By eliminating shaking and employing the soft delamination approach described in this report, it is possible to translate the size of the parent MAX particles into individual MXene flakes. By using large MAX particles as precursors, we demonstrate MXene flakes of up to 40 μm in size. These flakes can be used for fabrication of various electronic devices with enough space for patterning multiple contacts on a single flake. We anticipate the results of this work will further facilitate investigation of fundamental properties of MXenes at the individual flake level and will initiate fabrication of MXene-based materials with improved properties due to the increased flake size.

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EXPERIMENTAL SECTION

Synthesis of the Ti_3AlC_2 MAX Phase. Synthesis of the Ti_3AlC_2 MAX phase was adapted from ref 26. In short, TiC, Ti, and Al powders (2:1:1 mass ratio) were ball-milled for 18 h using zirconia balls. The mixture was annealed at 1400 °C for 2 h under an Ar atmosphere. The sintered compact was powdered and washed with 9 M HCl for 18 h to remove the intermetallic impurities. The resulting MAX powder was washed with water and dried under vacuum at 80 °C for 6 h and sieved using 400 mesh sieves to obtain MAX particles with a particle size below 38 μm.

To fabricate large MAX particles, the ball-milled precursor powders (TiC, Al, Ti) were annealed at 1400 °C for 24 h. The sintered block (without grinding or powdering) was immersed in 9 M HCl for 3 days to remove the intermetallic impurities, then washed with water until neutral pH and dried in a vacuum at 80 °C for 6 h. The resulting powders were separated into three samples with particles of <20, 20−38, and 38−75 μm.

Etching $Ti_3C_2T_x$ MXene. Synthesis of $Ti_3C_2T_x$ MXene was adapted from ref 26. One gram of Ti_3AlC_2 MAX powder was slowly added to 20 mL of the mixture of concentrated HF (49 wt %), concentrated HCl (36 wt %), and DI H_2O with a volumetric ratio of 1:6:3. The MAX phase was etched at 35 °C for a specific amount of time, which was varied between 24 and 72 h. After the reaction was complete, MXene was washed with DI water *via* multiple cycles of centrifugation until neutral pH.

Delamination of $Ti_3C_2T_x$ MXene. To intercalate Li^+ ions between the layers of MXene, the multilayer MXene was stirred at 300 rpm in a solution of 1 g of LiCl in 50 mL of water at 35 °C for 18 h. For the regular delamination, LiCl was removed from the solution by centrifuging it for 5 min at 3500 rpm and discarding the clear supernatant. After that, the MXene powder was completely redispersed in water *via* shaking and centrifuged for 15 min at 3500 rpm (to precipitate the multilayer MXene and unreacted MAX particles), and the dark supernatant containing single-layer MXene was collected.

For the soft delamination, after the LiCl intercalation, the solution was left undisturbed for approximately 30 min to precipitate the multilayer MXene particles, and the clear supernatant was removed *via* pipetting. Then the vial was refilled with water, MXene was gently redispersed by inverting the vial several times, and the vial was again left undisturbed for another 30 min, after which the supernatant was collected. After several cycles, the MXene started to delaminate spontaneously, and the solution became dark. We continued collecting the supernatant following these steps until all the dark solution was collected.

Characterization of MXene. MXene solutions were drop-casted on Si/SiO_2 substrates with a 300 nm oxide layer and imaged *via* a KEYENCE VK-X1000 optical microscope. AFM analysis was performed using a Bruker Multimode 8 instrument in tapping mode. SEM analysis was performed using a Zeiss Supra 50VP electron microscope. TEM imaging was performed using a JEOL 2100 instrument. A Rigaku Miniflex diffractometer was used for XRD studies with Cu Kα radiation and the following conditions: (i) for the MAX powder, step scan of 0.02°, 2θ = 5−90°, step time of 1 s; (ii) for the MXene films, step scan of 0.03°, 2θ = 3−70°, step time of 0.5 s.

Electrical Characterization of Individual MXene Flakes. $Ti_3C_2T_x$ flakes fabricated from the regular MAX powder (annealed for 2 h) *via* soft delamination protocol were deposited on Si/SiO_2 substrates and then used for device fabrication, as described in our previous work.[11] In brief, the contacts on $Ti_3C_2T_x$ flakes were patterned by electron beam lithography using a Zeiss Supra 40 field-emission scanning electron microscope and a Raith pattern generator, and then the metallization of contacts was performed using an AJA electron beam evaporation system. The metallization involved a deposition of 3 nm of Cr and 15 nm of Au. The $Ti_3C_2T_x$ devices were imaged by SEM using a Zeiss Supra 40 field-emission scanning electron microscope, as well as by AFM using a Bruker Dimension Icon atomic force microscope. The electrical measurements were performed using a Lake Shore TTPX cryogenic probe station, which was operated at a base pressure of about 2×10^{-6} Torr, and an Agilent 4155C semiconductor parameter analyzer.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.2c04506.

Figures of electron microscopy characterization of the Ti_3AlC_2 MAX phase and the large MXene flakes, size distribution for the MAX phase particles used in this study and the single-layer MXene flakes fabricated *via* soft delamination, optical microscopy of the large MXene flakes deposited *via* spin-coating at different speeds, *in situ* optical microscopy of an aged sample upon drying, optical microscopy of the large flakes fabricated from the 20−38 μm fraction of the MAX phase etched for 72 h and delaminated using the soft delamination approach (PDF)

Supplementary Video 1: Delamination of MXene flakes in solution upon water evaporation tracked *via* optical microscopy (MP4)

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Notes

The authors declare no competing financial interest.

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